

5. The unit-impulse response can be used to fully describe a linear, time-invariant system. Given the impulse response, the system output is the sum of the products of the impulse response coefficients and corresponding input samples, called the digital convolution sum.
6. BIBO is a type of stability in which a bounded input will produce a bounded output. A BIBO system requires that the sum of the absolute impulse response coefficients be a finite number.
7. The digital convolution sum, which represents a DSP system, is evaluated in three ways: the graphical method, evaluation of the formula, and the table method. The table method is found to be most effective.

3.7 PROBLEMS

3.1. Sketch each of the following special digital sequences:

- a. $5\delta(n)$
- b. $-2\delta(n-5)$
- c. $-5u(n)$
- d. $5u(n-2)$

3.2. Calculate the first eight sample values and sketch each of the following sequences:

- a. $x(n) = 0.5^n u(n)$
- b. $x(n) = 5\sin(0.2\pi n)u(n)$
- c. $x(n) = 5\cos(0.1\pi n + 30^\circ)u(n)$
- d. $x(n) = 5(0.75)^n \sin(0.1\pi n)u(n)$

3.3. Sketch each of the following special digital sequences:

- a. $8\delta(n)$
- b. $-3.5\delta(n-4)$
- c. $4.5u(n)$
- d. $-6u(n-3)$

3.4. Calculate the first eight sample values and sketch each of the following sequences:

- a. $x(n) = 0.25^n u(n)$
- b. $x(n) = 3\sin(0.4\pi n)u(n)$
- c. $x(n) = 6\cos(0.2\pi n + 30^\circ)u(n)$
- d. $x(n) = 4(0.5)^n \sin(0.1\pi n)u(n)$

3.5. Sketch the following sequences:

- a. $x(n) = 3\delta(n+2) - 0.5\delta(n) + 5\delta(n-1) - 4\delta(n-5)$
- b. $x(n) = \delta(n+1) - 2\delta(n-1) + 5u(n-4)$

- 3.6. Given the digital signals $x(n]$ in Figures 3.24 and 3.25, write an expression for each digital signal using the unit-impulse sequence and its shifted sequences.

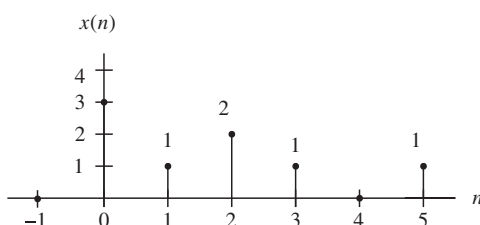


FIGURE 3.24

The first digital signal in Problem 3.6.

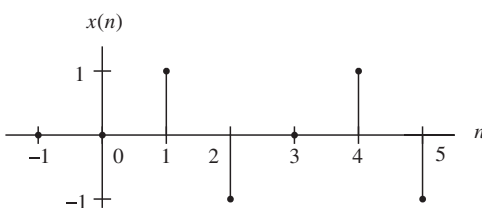


FIGURE 3.25

The second digital signal in Problem 3.6.

- 3.7. Sketch the following sequences:

- $x[n] = 2\delta[n + 3] - 0.5\delta[n + 1] - 5\delta[n - 2] - 4\delta[n - 5]$
- $x[n] = 2\delta[n + 2] - 2\delta[n + 1] + 5u[n - 3]$

- 3.8. Given the digital signals $x[n]$ in Figures 3.26 and 3.27, write an expression for each digital signal using the unit-impulse sequence and its shifted sequences.



FIGURE 3.26

The first digital signal in Problem 3.8.

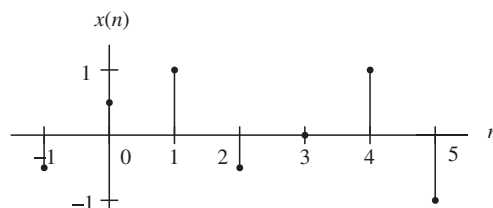


FIGURE 3.27

The second digital signal in Problem 3.8.

- 3.9.** Assume that a DS processor with a sampling time interval of 0.01 second converts the following analog signals $x(t)$ to a digital signal $x(n)$; determine the digital sequence for each of the analog signals.
- $x(t) = e^{-50t}u(t)$
 - $x(t) = 5\sin(20\pi t)u(t)$
 - $x(t) = 10\cos(40\pi t + 30^\circ)u(t)$
 - $x(t) = 10e^{-100t}\sin(15\pi t)u(t)$
- 3.10.** Determine which of the following systems is a linear system.
- $y(n) = 5x(n) + 2x^2(n)$
 - $y(n) = x(n-1) + 4x(n)$
 - $y(n) = 4x^3(n-1) - 2x(n)$
- 3.11.** Assume that a DS processor with a sampling time interval of 0.005 second converts each of the following analog signals $x(t)$ to a digital signal $x(n)$; determine the digital sequence for each of the analog signals.
- $x(t) = e^{-100t}u(t)$
 - $x(t) = 4\sin(60\pi t)u(t)$
 - $x(t) = 7.5\cos(20\pi t + 60^\circ)u(t)$
 - $x(t) = 20e^{-200t}\sin(60\pi t)u(t)$
- 3.12.** Determine which of the following systems is a linear system.
- $y(n) = 4x(n) + 8x^3(n)$
 - $y(n) = x(n-3) + 3x(n)$
 - $y(n) = 5x^2(n-1) - 3x(n)$
- 3.13.** Determine which of the following linear systems is time invariant.
- $y(n) = -5x(n-10)$
 - $y(n) = 4x(n^2)$

- 3.14.** Determine which of the following linear systems is causal.
- $y(n) = 0.5x(n) + 100x(n-2) - 20x(n-10)$
 - $y(n) = x(n+4) + 0.5x(n) - 2x(n-2)$
- 3.15.** Determine the causality for each of the following linear systems.
- $y(n) = 0.5x(n) + 20x(n-2) - 0.1y(n-1)$
 - $y(n) = x(n+2) - 0.4y(n-1)$
 - $y(n) = x(n-1) + 0.5y(n+2)$
- 3.16.** Find the unit-impulse response for each of the following linear systems.
- $y(n) = 0.5x(n) - 0.5x(n-2)$; for $n \geq 0$, $x(-2) = 0$, $x(-1) = 0$
 - $y(n) = 0.75y(n-1) + x(n)$; for $n \geq 0$, $y(-1) = 0$
 - $y(n) = -0.8y(n-1) + x(n-1)$; for $n \geq 0$, $x(-1) = 0$, $y(-1) = 0$
- 3.17.** Determine the causality for each of the following linear systems.
- $y(n) = 5x(n) + 10x(n-4) - 0.1y(n-5)$
 - $y(n) = 2x(n+2) - 0.2y(n-2)$
 - $y(n) = 0.1x(n+1) + 0.5y(n+2)$
- 3.18.** Find the unit-impulse response for each of the following linear systems.
- $y(n) = 0.2x(n) - 0.3x(n-2)$; for $n \geq 0$, $x(-2) = 0$, $x(-1) = 0$
 - $y(n) = 0.5y(n-1) + 0.5x(n)$; for $n \geq 0$, $y(-1) = 0$
 - $y(n) = -0.6y(n-1) - x(n-1)$; for $n \geq 0$, $x(-1) = 0$, $y(-1) = 0$
- 3.19.** For each of the following linear systems, find the unit-impulse response, and draw the block diagram.
- $y(n) = 5x(n-10)$
 - $y(n) = x(n) + 0.5x(n-1)$
- 3.20.** Determine the stability of the following linear system.
- $$y(n) = 0.5x(n) + 100x(n-2) - 20x(n-10)$$
- 3.21.** For each of the following linear systems, find the unit-impulse response, and draw the block diagram.
- $y(n) = 2.5x(n-5)$
 - $y(n) = 2x(n) + 1.2x(n-1)$
- 3.22.** Determine the stability for the following linear system.
- $$y(n) = 5x(n) + 30x(n-3) - 10x(n-20)$$
- 3.23.** Determine the stability for each of the following linear systems.
- $y(n) = \sum_{k=0}^{\infty} 0.75^k x(n-k)$
 - $y(n) = \sum_{k=0}^{\infty} 2^k x(n-k)$

3.24. Determine the stability for each of the following linear systems.

a. $y(n) = \sum_{k=0}^{\infty} (-1.5)^k x(n-k)$

b. $y(n) = \sum_{k=0}^{\infty} (-0.5)^k x(n-k)$

3.25. Given the sequence

$$h(k) = \begin{cases} 2, & k = 0, 1, 2 \\ 1, & k = 3, 4 \\ 0 & \text{elsewhere,} \end{cases}$$

where k is the time index or sample number,

a. sketch the sequence $h(k)$ and the reverse sequence $h(-k)$;

b. sketch the shifted sequences $h(-k+2)$ and $h(-k-3)$.

3.26. Given the sequence

$$h(k) = \begin{cases} -1 & k = 0, 1 \\ 2 & k = 2, 3 \\ -2 & k = 4 \\ 0 & \text{elsewhere} \end{cases}$$

where k is the time index or sample number,

a. sketch the sequence $h(k)$ and the reverse sequence $h(-k)$;

b. sketch the shifted sequences $h(-k+1)$ and $h(-k-2)$.

3.27. Using the sequence definitions

$$h(k) = \begin{cases} 2, & k = 0, 1, 2 \\ 1, & k = 3, 4 \\ 0 & \text{elsewhere} \end{cases} \quad \text{and} \quad x(k) = \begin{cases} 2, & k = 0 \\ 1, & k = 1, 2 \\ 0 & \text{elsewhere} \end{cases}$$

evaluate the digital convolution

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

a. using the graphical method;

b. using the table method;

c. applying the convolution formula directly.

3.28. Using the sequence definitions

$$x(k) = \begin{cases} -2, & k = 0, 1, 2 \\ 1, & k = 3, 4 \\ 0 & \text{elsewhere} \end{cases} \quad \text{and} \quad h(k) = \begin{cases} 2, & k = 0 \\ -1, & k = 1, 2 \\ 0 & \text{elsewhere} \end{cases}$$

evaluate the digital convolution

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

- a. using the graphical method;
- b. using the table method;
- c. applying the convolution formula directly.

3.29. Convolve the two rectangular sequences

$$x(n) = \begin{cases} 1 & n = 0, 1 \\ 0 & \text{otherwise} \end{cases} \text{ and } h(n) = \begin{cases} 0 & n = 0 \\ 1 & n = 1, 2 \\ 0 & \text{otherwise} \end{cases}$$

using the table method.